MGU-BVOC (HONOURS) REGULAR EXAMINATION, NOVEMBER 2025

FIRST SEMESTER

MDC- MG1MDCAFP100- READY TO EAT CONVENIENCE FOOD

QP CODE: 25901013

Part – A

Short Answer

Answer any 5. Each carry 2 marks.

1. List out the types of RTD beverages?
   RTD Tea & Coffee, Dairy-Based RTD Drinks, Fruit & Vegetable Juices, Functional RTD Drinks, Meal Replacement Drinks, Alcoholic RTD Drinks
2. Which are the pre-processing steps for raw materials?
   - **Sorting and grading:** To ensure uniformity.
   - **Washing and sanitizing:** To remove dirt and microorganisms.
   - **Peeling, cutting, and trimming:** For processing convenience.
   - **Blanching or pre-cooking:** To inactivate enzymes and reduce microbial load.
   - **Weighing and mixing:** To maintain formulation consistency.
3. What is the purpose of allergen management in food safety?
   Allergen Management is a critical component of food safety systems such as HACCP, ISO 22000, FSSAI, and FDA FSMA. It ensures that foods are safe for consumers with food allergies and helps prevent allergic reactions caused by unintended allergen presence.
4. What leads to the evolution of packaged foods in 20th century?
   - Refrigeration and pasteurization allowed perishable foods to last longer.
   - The rise of food brands introduced standardized, packaged products.
   - World Wars I & II (1914–1945) demanded compact, durable, nutritious foods for soldiers. After the wars, these military technologies were adapted for civilian markets, giving rise to the instant food industry.
5. List out the functions of FSSAI?
   - Lays down science-based food standards.
   - Regulates the manufacture, storage, distribution, sale, and import of food.
   - Ensures availability of safe and wholesome food to consumers.
   - Acts as a *single reference point* for all food-related laws in India.
6. List out some packaging materials used in RTE?
   Plastics (Flexible and Rigid), Glass containers, Metal cans (Tin / Aluminium), Paper and paperboard, Laminates and composites
7. Name some common additives used in RTE foods?
   Antioxidants- BHA, BHT
   Emulsifiers- lecithin
   Flavouring agents- natural and artificial
   Colouring agents- caramel, sunset yellow

Part-B

Short Essay

Answer any 5. Each carry 5 marks.

8. What are the mandatory information to be given on a food label?
   - Name of the food
   - List of ingredients
   - Nutritional information
   - Net quantity (weight/volume)
   - Manufacturer/packer details
   - Batch/Lot number
   - Date of manufacture & expiry / best-before
   - Veg/Non-veg symbol
   - MRP (Maximum Retail Price)
   - Instructions for use/storage
9. Write about non-thermal processing and its objectives?

   Non-thermal processing involves methods that preserve food without the use of high heat, thus retaining nutritional, sensory, and functional qualities while still ensuring microbial safety.

   Objectives:
   - To inactivate microorganisms without significant heating.
   - To retain natural taste, colour, and nutrients.
   - To reduce energy consumption compared to thermal methods.
10. Write a note on the characteristics and processing of RTS?

    Characteristics
    - Require no further processing — consumed directly from the pack.
    - Have stable flavor, color, and texture during storage.
    - Are nutritionally balanced, often enriched with vitamins or minerals.
    - Have extended shelf life due to thermal or non-thermal processing (like pasteurization, UHT, or aseptic packaging).

    Processing Steps
    1. Selection of raw materials – high-quality fruits, milk, or vegetables.
    2. Extraction or preparation – juice extraction, blending, or homogenization.
    3. Formulation – adjusting sugar, acidity (using citric acid), and flavoring.
    4. Filtration and pasteurization – to remove impurities and destroy microbes.
    5. Filling and packaging – usually in glass bottles, Tetra Pak cartons, PET bottles, or cans.
    6. Cooling and storage – stored under ambient or refrigerated conditions.

11. Write a note on the classification of RTE foods?
    - Based on degree of preparation- RTE, RTC
    - Based on the method of processing- canned, frozen, dried, retort-packed, vaccum packed
    - Based on storage requirements- shelf-stable, chilled, frozen
    - Based on type of food product- cereal based, meat and fish, fruits and veg, traditional
    - Based on packaging type- canned, vaccum, retort pouch, MAP, frozen pack
    - Based on consumer use- single-meal packs, family packs, instant

12. Write a note on freeze drying and dehydration?
    Dehydration is the process of removing moisture from food by applying heat, usually by evaporation of water.
    Freeze drying is a low-temperature dehydration process in which frozen food is dried under vacuum, causing ice to sublimate (convert directly from solid to vapour) without melting.
13. Write about the methods of shelf-life assessment?
    Real-Time Testing, Accelerated Shelf-Life Testing (ASLT), Microbial Challenge Studies
14. Write a note on value addition and fortification of RTE foods?
    Value addition means improving a food's appeal, nutrition, or convenience.
    Fortification involves enriching foods with essential nutrients to enhance their health benefits.

Part- C

Essay

Answer any 1. Each carry 15 marks

15. Write an essay on FDA and ISO?
    The Food and Drug Administration (FDA) is a regulatory agency of the United States federal government responsible for safeguarding public health.
    It oversees the safety, quality, and effectiveness of:
    - Foods
    - Drugs (human & veterinary)
    - Cosmetics
    - Medical devices
    - Biological products
    - Tobacco products

- ISO (International Organization for Standardization) is an independent, non-governmental international body that develops and publishes voluntary global standards to ensure quality, safety, efficiency, and consistency in products, services, and systems.

16. Write an essay on RTD foods?
    RTD (Ready-to-Drink) beverages are pre-formulated, processed, and packaged drinks that are ready for direct consumption without any additional preparation like dilution, brewing, or mixing.

    ## Purpose

    The main goal of RTD products is convenience.
    They are designed for consumers who have busy lifestyles and want quick, nutritious, and safe beverage options on the go.

    ## Characteristics

    1. Shelf-stable – Many RTD drinks can be stored at room temperature due to aseptic or UHT (Ultra-High Temperature) processing.
    2. Portable – Available in easy-to-carry containers such as bottles, cans, or tetra packs.
    3. Ready for immediate consumption – No need for mixing, brewing, or refrigeration before drinking.
    4. Wide variety – Includes both health-based and indulgent options.
    5. Preserved flavor and nutrition – Processing methods are chosen to maintain taste and nutrients for a longer shelf life.

Advantages of RTD Beverages

1. Convenient and time-saving
2. Portable and hygienic packaging
3. Long shelf life
4. Consistent taste and quality
5. Suitable for all age groups and occasions

Limitations / Concerns

1. May contain **added sugar or preservatives**
2. Some may lose **nutritional value** during processing
3. **Costlier** than preparing homemade beverages
4. Environmental concerns related to **single-use packaging**